

AI FOR VALIDATION

CHAPTER 2

The Model Validation Framework

2

Agus Sudjianto and Gary Ang

Ground Truth Reference

Before searching for failure, define what counts as correct behavior.

1 PREDICTIVE REFERENCE

Observed outcomes

For predictive models, labeled observations are available.

Predictions are compared with the observed outcome.

2 GENAI / AGENTIC REFERENCE

Checked facts and policies

For GenAI and agents, the reference may be source-backed facts, checked answers or response policies.

Tool and action tests need rules for permitted behavior.

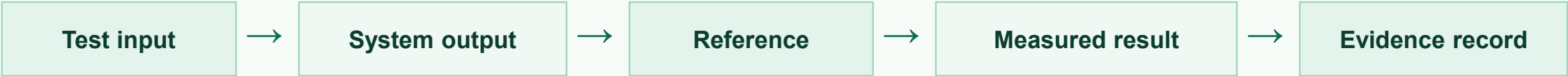
3 KEY PRINCIPLE

Verdict

A missing reference is not evidence of failure.

It is an evidence gap that must be recorded, repaired or scoped out.

4 REFERENCE TURNS TESTS INTO EVIDENCE



Find Conditional Weakness

Average performance can hide where the model or agent actually fails.

1 THE PROBLEM WITH AVERAGES

Overall performance can look acceptable

A global score summarizes the whole population or task set. It can miss small but material pockets of failure.

2 WHAT VALIDATION SEARCHES FOR

Failure under conditions

Conditional weakness means poor performance associated with a customer group, period, stress, prompt, tool state or interaction.

3 SIMPLE EXAMPLE



Key principle

Validation is not only “How well does the model perform?” but “Where, when and under what conditions does it fail?”

One-Shot Assessment: Useful but Limited

Initial evaluation can reveal weakness before any adaptive loop begins.

1 WHAT ONE-SHOT CAN DO

Predictive model

Partition test observations, then measure performance inside each group or cluster.

GenAI / agent

Run a designed set of tasks with known references or response policies.

Output

A table of measured weaknesses, failed cases, unevaluable cells and coverage gaps.

2 THE LIMITATION

One-shot only covers the regions, cases and conditions that were included in the initial design.

When the results point to a narrower failure, a missing condition or a better candidate test, the assessment naturally becomes adaptive discovery.

Inner Array × Outer Array

Separate what the search varies from where every candidate is evaluated.

1 INNER ARRAY

Candidate tests

What we search or vary:

- customer regions
- medoid sets
- task presentations
- prompt factors

2 OUTER ARRAY

Evaluation conditions

Where the candidate is challenged:

- samples and vintages
- stress regimes
- source access
- tool faults

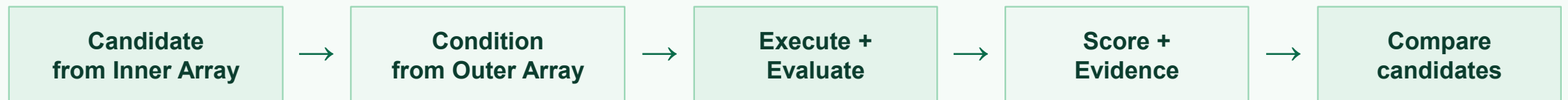
3 SCORE TABLE

Comparable measurements

Crossing candidates with conditions produces scores and evidence records.

Missing or failed cells remain visible.

4 HOW TO READ THE FRAMEWORK



From Discovery to Confirmed Findings

Search tells us where to look; confirmation tells us what we can conclude.

1 ADAPTIVE SEARCH



2 WHAT SEARCH ESTABLISHES

Discovery score

A high discovery score identifies a promising weakness candidate. It does not by itself establish a finding.

3 WHAT CONFIRMATION ESTABLISHES

Reserved evidence

The selected test is frozen, then assessed on evidence that did not guide the adaptive search. (held out samples)

Key principle

Discovery is for investigation. Confirmation is for conclusions.
